

AMMAR HASHIM

Electronics Engineer

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PROFESSIONAL SUMMARY

MSc Electronics & Information Technology graduate with hands-on experience in satellite communications, FPGA-based digital design, SDR systems, signal processing, and power electronics. Skilled in VHDL, Verilog, GNU Radio Companion, MATLAB/Simulink, and communication system analysis through practical engineering projects.

TECHNICAL SKILLS

- Communication Systems: Satellite Communications, SDR Systems, RF Systems, Signal Processing
- FPGA & Digital Design: VHDL, Verilog, FPGA Design, Vivado
- Programming: C, C++, MATLAB, Simulink
- Power Electronics: Boost Converters, Wireless EV Charging
- Microcontrollers & IoT: Arduino, ATmega, Sensor Interfacing
- Software & Tools: GNU Radio Companion, SDR++, LTspice, Proteus, Atmel Studio, LabVIEW, STK

EDUCATION

MSc Electronics & Information Technology – University of South Wales, UK (2025)

BEng Electrical Engineering – COMSATS University Islamabad, Pakistan (2022)

PROJECTS & RESEARCH

SDR-Based Experiments for Future 6G Frequencies

Designed SDR-based communication systems using HackRF One and GNU Radio Companion for transmission and reception of text, image, and audio data.

Satellite Constellation Design for Optimal Coverage

Designed multi-satellite constellation coverage for Cardiff, Perth, and Durban across S, UHF, and VHF bands using STK.

FPGA-Based Electrical Energy Monitoring System

Developed a Verilog-based FPGA energy monitoring system with GSM communication and cloud integration.

8-Bit Configurable Timer Peripheral

Designed a configurable timer peripheral supporting one-shot and repeat modes with auto-reload functionality.

Hydrogen Fuel Cell with Boost Converter

Simulated hydrogen fuel cell systems in Simulink with Boost and Cuk converter analysis.

Dynamic Wireless Charging System for Electric Vehicles

Designed a 2 kW wireless EV charging system operating at 50 kHz for inductive power transfer.

PROFESSIONAL EXPERIENCE

Technical Support Engineer – PTCL, Islamabad (Mar 2023 – Dec 2023)

- Configured and troubleshot IPTV, broadband, and telephony systems.
- Worked with Huawei NMS, Nokia AMS, and ZTE network management systems.
- Diagnosed GPON, ADSL, and VDSL network faults.

ADDITIONAL INFORMATION

- Eligible to obtain UK Security Clearance and hold UK Driving Licence.
- Strong interests in FPGA systems, satellite communications, SDR research, and power electronics.